UWB TDoA Real-time Localization Proposal

EC601 course project | 2 October 2026 | Proposed design and validation plan

I propose a laboratory prototype that estimates the positions of two tagged objects using three fixed ultra wideband anchors and a central computer. The primary technical objective is to demonstrate uplink time difference of arrival (TDoA) localization. The recommended first application is equipment location; human and robot tracking remain later application hypotheses. This is a one-person project; I am responsible for hardware, firmware, host software, evaluation and documentation, with AI assistance.

Evidence status: No prototype results, user interviews, inventory, budget, available development time, or final course deadline have been verified. External facts carry source IDs. All design choices, numerical acceptance criteria, personas, and schedules below are proposals, not achieved results.

Five Ws and How

What — Build a system that shows where two tagged objects are on a small indoor map and indicates when a position is unreliable.

Who — Serve an equipment coordinator first; I operate the setup as the sole developer, course staff assess evidence, and tagged people would be affected by later human tracking.

Why — Test whether measured tag positions help users find equipment; current search delays and the benefit over their existing workflow must be established through interviews.

When — Use it during a supervised equipment-location demo; target a full demo after four proposed one-week sprints, subject to the actual course schedule.

Where — Begin in a surveyed laboratory area with clear radio paths and known tag height; warehouse and factory deployment require separate validation.

How — Tags send UWB broadcasts; anchors record hardware arrival timestamps; clock correction and Wi-Fi forwarding feed a computer that estimates positions and displays quality.

Mission statement

For an equipment coordinator who needs to find tagged equipment in a defined indoor area, my UWB localization prototype is a location display that provides measured positions and freshness information; unlike a manual location log, it aims to update positions automatically and expose the evidence behind each estimate.

The need and claimed advantage are hypotheses. A stranger can reject the mission if users do not need position updates or if the measured locations are too inaccurate to be useful. The proposal follows the supplied guide’s definition, mission, users, stories, slicing, and validation sequence [G1].

Technical basis and architecture

Verified facts and their limits

Qorvo lists the DWM3000 for TWR and TDoA tag or anchor use, with a customer-supplied microcontroller and no integrated Bluetooth. Asset tracking and factory or warehouse automation are listed application areas. These listings support component selection, not a promise about this prototype's accuracy, capacity, or battery life [S1].

A published ESP32 and DWM3000 prototype demonstrates the hardware pairing using TWR and five anchors. Its Table II reports position-dependent mean errors from 9.27 cm to 121.37 cm. Its abstract's "up to 10 cm" statement is therefore insufficient evidence for uniform 10 cm accuracy, and its TWR results do not validate my TDoA design [S2].

The classic ESP32 series provides Wi-Fi, Bluetooth/BLE and SPI. The exact MCU, development board, supported UWB driver, pin mapping, and power supply must be selected and tested; capabilities should be checked for that specific part rather than assumed across the ESP32 family [S3].

Proposed data path

1. Two tags transmit UWB frames containing tag ID and sequence number at scheduled, separate times.

2. Three anchors receive the same frame and read UWB hardware timestamps through the MCU. MCU interrupt time is not used as the radio arrival measurement.

3. A synchronization mechanism maps each anchor clock to a common time basis. The first candidate is periodic UWB reference messages with correction for known anchor separation and clock offset/drift. Its feasibility is unproven.

4. Anchors forward tag ID, sequence, anchor ID, raw timestamp, clock-calibration version, and available receive-quality diagnostics over Wi-Fi. The computer groups observations from the same frame.

5. The computer corrects clocks, calculates arrival-time differences, solves for position, rejects invalid results, and displays coordinates, age, and quality. Raw records remain available for replay.

Timing and geometry

In uplink TDoA, the measurement is a difference in arrival times, corresponding to a difference in path lengths; it is not an independently measured tag-to-anchor distance. Clock correction is essential [S2, S4]. For anchors i and 1, the ideal model is $c \times (t_i - t_1)$ = distance(tag, anchor i) − distance(tag, anchor 1).

Engineering implication: Three anchors provide two independent differences for two horizontal unknowns when height is known. They offer no extra measurement redundancy and do not guarantee a unique or well-conditioned solution. Constrain the surveyed region and reject ambiguous solutions. Recommend a fourth anchor if resources allow. Radio placement and obstructed paths affect TDoA accuracy [S4].

Calculated timing scale: Using the speed of light, approximately 3×10^8 m/s [S5], a 1 ns timing error equals about 0.30 m of path-difference error before geometric amplification. Wi-Fi/BLE transport must preserve radio timestamps; computer packet-arrival times cannot replace them.

Users and project scope

Primary user and subproducts

The primary user is the warehouse equipment coordinator, a role to recruit and validate rather than an invented named customer. Proposed persona: they look for tagged carts or tools, need a readable map and last-seen indication, and should be able to tell when the displayed position is stale. Actual workflow, search time, acceptable error, and willingness to attach tags are unknown.

Role	Subproduct	Proposed measure
Equipment coordinator	Live map with tag identity and freshness	Find the correct zone for both demo tags; interview usefulness separately
System developer	Anchor setup and diagnostic log	Repeat setup and replay a recorded session
Course assessor	Evaluation report with raw evidence	Reproduce reported error and packet-loss metrics
Robot developer or tagged person	Future position API or wearer controls	Deferred until the core demo and use case are validated

The deployment decider would be a site operations or engineering manager; none has been identified. Workers are stakeholders if tags are worn or movement is inferred. The first demo will use equipment tags and pseudonymous IDs. Human tracking requires explicit participant agreement and a separate decision on access and retention; this is a proposed project constraint.

Scope and alternatives

Choice	Reason and limitation
Initial scope	Two tags, three anchors, known height, measured indoor area, central solver, live map and replayable records.
TDoA objective	Keep uplink TDoA as the research goal; make clock synchronization a first-sprint feasibility decision.
TWR baseline	Use briefly to check hardware or as an explicit fallback if synchronization fails. A successful TWR demo must be labeled TWR [S2].
Backhaul	Choose Wi-Fi first to simplify the central-computer path. Keep USB logging as a bench diagnostic; add BLE only for a demonstrated need.
Later scope	Human tracking, robot control, free-height 3D, industrial coverage, large tag counts, custom PCB and production battery claims require further evidence.

Better framing: Position this project as a measured TDoA feasibility demonstration, rather than a general factory tracking product. With only two tags, a scalability advantage remains untested. For such a small demonstration, TWR is a credible fallback; TDoA is justified by the learning objective and potential later expansion, not by measured superiority.

User validation before implementation

Interview three prospective users about the last time they searched for equipment, their current process, the smallest useful location zone, and situations where an incorrect location would be worse than no location. Test a paper map with freshness labels. Do not assume that a new dashboard fits their workflow. Record anonymized findings and revise the primary persona and mission [G1].

Five user stories and acceptance criteria

These are proposed acceptance criteria for the course prototype. They need user and course confirmation. Technical experiments support the stories; they are not substitutes for user-visible outcomes. Priority follows the risk that could invalidate the demonstration [G1].

Story 1 A position with traceable measurements

As the equipment coordinator, I want to see the location of one tag at marked positions, so that I can judge whether it points to the right place.

Acceptance: At nine surveyed positions spanning the chosen area, collect 100 scheduled transmissions per position. Retain matched observations from all three anchors. For valid estimates, meet the proposed median horizontal error of 0.30 m and 95th-percentile error of 0.75 m; obtain a valid estimate for at least 95% of scheduled transmissions. Report both failures and errors, with links from map points to underlying records.

First dependency: demonstrate clock correction on held-out data. If that fails, revise the approach before polishing the map. Timing criteria alone do not guarantee position criteria.

Story 2 Two distinct moving tags

As the equipment coordinator, I want two tags to remain separately identifiable as they move, so that I do not confuse one object with another.

Acceptance: Run two tags for ten minutes at a proposed 10 transmissions per second each. Show distinct IDs with no identity swaps. For each tag, produce valid positions for at least 95% of scheduled transmissions. Measure achieved update intervals; do not infer them from configured transmit rate. Compare with a single-tag run and report collisions or missing observations.

Story 3 A clear stale or invalid state

As the equipment coordinator, I want the display to tell me when location evidence is missing or invalid, so that I do not act on an old position as if it were current.

Acceptance: Interrupt an anchor link or remove a tag. Within one second of the last accepted estimate, mark the position stale and display its age. Never generate a new position from fewer than the required matched anchor observations or from invalid clock calibration. Preserve the last good point with an explicit stale label.

Story 4 Repeatable setup

As the system developer, I want to load a surveyed anchor configuration and run a startup check, so that a reviewer can reproduce the demo.

Acceptance: A reviewer follows the written instructions without editing source code, selects the coordinate frame and tag height, loads anchor coordinates, and obtains a checked estimate. Startup rejects duplicate anchor IDs, missing coordinates, and degenerate anchor geometry. Record firmware and configuration versions.

Story 5 Reproducible results

As the course assessor, I want the raw session and an evaluation script, so that I can verify the claimed accuracy and limitations.

Acceptance: Replay a saved session to reproduce accepted/rejected counts and reported error statistics within declared numerical tolerance. Include ground-truth points, packet sequences, calibration records, solver version, and an obstructed-path experiment. Label measured results, failed criteria, proposed targets, and TWR fallback runs separately.

INVEST review and thin demo slices

Review of the five stories

Check	Application to this proposal
Independent	Map, freshness and replay can start with recorded data. Live position stories share clock and radio prerequisites; dependency is explicit rather than claiming full independence.
Negotiable	Preserve user outcomes while allowing MCU, synchronization, solver and transport choices to change after the feasibility spike.
Valuable	Each story ends in a map, understandable failure state, repeatable setup, or auditable result. Driver setup is a supporting task.
Estimable	Time-box driver and synchronization experiments before estimating live TDoA work. My available development time and hardware access are still unknown.
Small	Split Story 1 into recorded replay and one live fixed point before testing the entire area. Split Story 2 into fixed tags and then moving tags.
Testable	Use the explicit data counts, pass thresholds and failure tests in the story acceptance criteria. Someone other than the implementer reruns them.

This applies the guide's INVEST lens to the actual backlog; it does not imply that every story is already small or independent [G1].

End to end slices

Slice	Small demonstrable outcome
1 Recorded data	A clearly labeled recorded or simulated observation passes through matching, solver, map and replay. This validates integration, not UWB feasibility.
2 One live point	One real tag and three synchronized anchors produce a checked position at a surveyed point through the chosen backhaul.
3 Measured area	The same path works across nine positions, with explicit error and missing-data statistics.
4 Two tags	Both tag IDs appear without swaps, with measured update rate and packet completeness.
5 Failure and replay	Dropout and obstruction are visible; a reviewer reproduces the session and metrics.

Standing definition of done

Every live slice retains hardware timestamps, packet IDs, units, coordinate frame, calibration version and raw records. Reject missing or inconsistent observations instead of inventing measurements. Tests cover timestamp wraparound, reordering, duplicates, clock reset, expired calibration and stale data. Simulated records are visibly labeled and excluded from measured hardware claims.

Use equipment tags for the initial demo. Any future human experiment must record participant agreement and a retention/access plan. Keep credentials out of the repository. These are proposed standing constraints, not claims of compliance or deployment readiness.

Assumptions and tests that could reject the project

Ranked qualitatively by impact and uncertainty. Every result is “not tested” as of this draft. Record results before upgrading any assumption to a fact [G1].

1 Anchor synchronization can support useful TDoA

Risk: Independent anchor clocks create false path differences. Test: Verify driver and board access first, then record one stationary tag with three anchors. Fit clock correction on calibration data and evaluate independent held-out points and times, including a restart. Proposed screening target: 95th-percentile path-difference residual no greater than 0.15 m after known geometric correction. This is about 0.5 ns and is only a feasibility screen.

Pivot criterion: If two time-boxed one-day experiments cannot obtain consistent, replayable correction, investigate a supported development platform or board-compatible wired timing design. If still infeasible within the course, explicitly change scope to TWR and report why. Do not assume a shared ESP32 GPIO pulse alone solves UWB clock drift.

2 Three anchors and known height cover the area

Risk: Geometry, ambiguous solutions and obstructed paths defeat the desired accuracy. Radio placement and NLOS bias affect TDoA localization [S4]. Test: Evaluate several anchor arrangements and nine ground-truth positions, including boundary locations; use a coarse simulated geometry check before physical mounting.

Pivot criterion: If the spatial error or valid-position target fails after one documented calibration/placement revision, add a fourth anchor, reduce the declared area, or relax the target with justification. Reject any claimed full-area result that excludes difficult points without disclosure.

3 The selected boards and driver are practical

Risk: A DWM3000 and a nominally capable MCU do not constitute working firmware. Test: Confirm the actual board schematic, SPI/interrupt wiring, available timestamp and scheduled-transmit APIs, driver license, and build path; send and receive timestamped frames between two boards.

Pivot criterion: If basic timestamped communication cannot be reproduced in one day after receipt of the boards, switch to a supported evaluation platform rather than begin a custom PCB. Hardware pairing has precedent, but my software integration remains unverified [S2].

4 Two tags and backhaul retain enough matched frames

Risk: Radio collisions, packet loss or clock-calibration gaps reduce usable updates. Test: Compare one-tag and two-tag runs, then introduce delayed, reordered and dropped backhaul records. Match by IDs and sequences, not arrival order.

Pivot criterion: If either tag falls below the proposed 95% valid-position ratio, revise scheduling or buffering and rerun once. Reduce and disclose update rate if necessary; never conceal missing frames through smoothing.

5 A real user values the result

Risk: The display does not solve the coordinator’s actual task. Test: Interview three users and test the paper-map workflow before implementation. Pivot criterion: If no interviewee identifies a concrete equipment-search need and an acceptable location tolerance, revise the user and mission. Do not claim productivity savings without a baseline and a later comparison.

Evaluation plan and decision criteria

All numbers in this section are recommended prototype targets. I should confirm them after the first interviews and timing spike. No accuracy, reliability, latency, or battery target has been achieved.

Measure	Proposed test and report
Horizontal accuracy	Nine surveyed points, 100 scheduled transmissions each. Calculate error to physical ground truth for every valid estimate; report median, 95th percentile, maximum and per-point statistics. Target median ≤0.30 m and P95 ≤0.75 m.
Valid-position ratio	Valid complete estimates / scheduled tag transmissions. Proposed target ≥95% per tag. Count radio losses and solver rejection; Wi-Fi packet success alone is insufficient.
Update behavior	Configure 10 Hz per tag initially. Report achieved valid updates per second and the distribution of gaps, including two-tag operation.
Display delay	Target P95 ≤250 ms from receipt of the final required record at the computer to display update. Measure this with one host clock. Report RF-to-host transport delay separately only with a validated clock mapping.
Stale display	After stopping tag or anchor traffic, stale label appears within 1 s of the last accepted estimate. Show age and preserve last good position explicitly.
Obstructed paths	Add a documented obstruction and change tag orientation. Compare errors and valid-position ratio with LOS runs; this is characterization, not a guarantee of industrial performance.
Repeatability	Repeat after restart and have a reviewer replay the saved session. Preserve configuration, firmware, calibration and evaluator versions.

Measurement discipline

Proposed setup: Define a local coordinate frame, survey anchor positions and tag test points, and record tag height. Aim for ground-truth uncertainty no worse than 0.05 m and report the method and achieved uncertainty. Keep tag height fixed or measured; the 2D solver uses the actual 3D anchor coordinates to calculate slant path lengths.

Keep calibration points separate from evaluation points. Freeze solver and calibration settings before the main evaluation, then repeat if settings change. Report raw and filtered estimates separately because a smooth track can still be biased. A controlled lab result cannot establish factory or warehouse performance [S2, S4].

For moving tags, use a surveyed path to demonstrate continuity and identity. Claim dynamic position accuracy only if time-aligned ground truth is available. Otherwise report update behavior and tracking gaps, with that limitation stated.

Evidence to retain

Save raw observations, synchronization messages, transmit counters, calibration parameters, anchor survey, ground truth, software version, test conditions and rejected-record reasons. The report must distinguish attempted transmissions, received records, complete matched frames and accepted positions. Recompute results from retained data rather than screenshot readings.

Work plan resources and course deliverables

Proposed schedule

T0 is my planned start after hardware and course deadlines are confirmed. The one-week sprint duration below is a planning assumption. The supplied guide lists 23, 25 and 28 September 2026 milestones, all before this draft's date; it does not establish my current submission or final-demo deadline. Confirm the updated dates with course staff [G1].

Period	Outcome and decision
Before Sprint 1	Interview three users, test paper map, confirm board inventory, confirm solo workload and actual deadline.
Sprint 1	Run driver and clock-correction spikes, plus recorded-data map/replay. Decide whether uplink TDoA remains feasible.
Sprint 2	Deliver one live end-to-end TDoA position, then spatial-grid evaluation. Revise geometry or add an anchor if needed.
Sprint 3	Deliver two-tag operation, stale/invalid states and measured update behavior.
Sprint 4	Run obstruction and restart tests, have a reviewer reproduce results, and finalize demonstration and report.

Hardware and cost basis

Resource	Proposed quantity and status
UWB and MCU nodes	Five complete nodes: two tags and three anchors. DWM3000 plus selected compatible MCU/development board; ownership and availability unverified.
Optional fourth anchor	One additional complete node for redundancy or failed geometry tests.
Power and mounting	Power supplies, cables, safe tag batteries if used, anchor mounts and measured positions; quote after board selection.
Computer and network	One central computer and Wi-Fi access point; availability unverified.
Survey and diagnostics	Measuring equipment and, if needed, logic analyzer or oscilloscope; access unverified.

Budget is not yet priced. Obtain quotes for complete development boards and accessories rather than treating the bare UWB module price as system cost. Proposed total = five node costs + power/mounting + timing hardware if required + any survey/network costs; an optional sixth node is separate. Do not purchase a custom PCB until the driver and timing method pass.

Calculated backhaul example: For a chosen 64-byte observation record, three anchors × two tags × 10 frames/s × 64 bytes = 3,840 bytes/s (30.7 kbit/s) of observation payload. This excludes transport headers, synchronization and retries; it is a sizing example, not a measured throughput result or a reason to assume reliable delivery.

Submission and unresolved decisions

I own hardware, firmware, host software, testing and documentation. Publish the mission, user profile, stories, assumption results, software, raw data and evaluation report in the repository. Invite independent review of retained results. Confirm hardware, timing method, test area, targets, budget and deadline before implementation.

Sources and guide coverage

Sources were accessed on 2 October 2026. Source IDs support external factual claims; proposal targets and engineering recommendations are identified in the text. User needs, resource availability and test results require direct project evidence.

G1 EC601 Defining Your Project

Supplied course guide. Read all 12 pages, including the linked p7 page. Used for required structure and listed milestones, not as technical proof.

S1 Qorvo DWM3000 product information

Component integration, customer MCU, absent Bluetooth, TWR/TDoA roles and listed applications. Vendor capability statements do not establish this project's performance.

S2 Krebs and Herter 2024 ESP32 and DWM3000 positioning study

Sections II–VI, Table II and conclusion. TWR hardware precedent, measured static errors and synchronization limitation; not an uplink TDoA benchmark. arXiv preprint 2403.10194.

S3 Espressif ESP32 Series Datasheet

Radio and peripheral capabilities of the classic ESP32 series. Confirm the selected part and board separately.

S4 Zhao Goudar and Schoellig 2022 Finding the Right Place

TDoA geometry and obstacle-induced NLOS bias. IEEE Robotics and Automation Letters 7(3), 6075–6082; DOI 10.1109/LRA.2022.3165181. Supports placement and evaluation concerns, not a universal anchor-count guarantee.

S5 NIST Speed of light in vacuum

Defines c as exactly 299,792,458 m/s. Used for the approximate timing-to-path-difference calculations; indoor propagation and geometry add other errors.

Answers mapped to the supplied guide

Guide page	Answer in this proposal
1 Project definition	Five Ws, mission, users and backlog on pages 1–4.
2 Five Ws and How	Six explicit answers on page 1.
3 Mission	One positioning sentence and falsification test on page 1.
4 Multiple users	Primary user, subproducts, decider and stakeholders on page 3.
5 User stories	Five role/capability/outcome stories and acceptance criteria on page 4.
6 INVEST	Six checks and actual dependencies on page 5.
7 Thin slicing	End-to-end slices and standing constraints on page 5.
8 Major stories	Five prioritized stories on page 4; timing risk tested first.
9 Validate first	Five ranked assumptions, tests and pivot criteria on page 6; evaluation on page 7. Results remain untested.
10–12 Closing and dates	Schedule, dated-milestone caveat and repository deliverables on page 8. Divider and thank-you pages add no substantive questions.